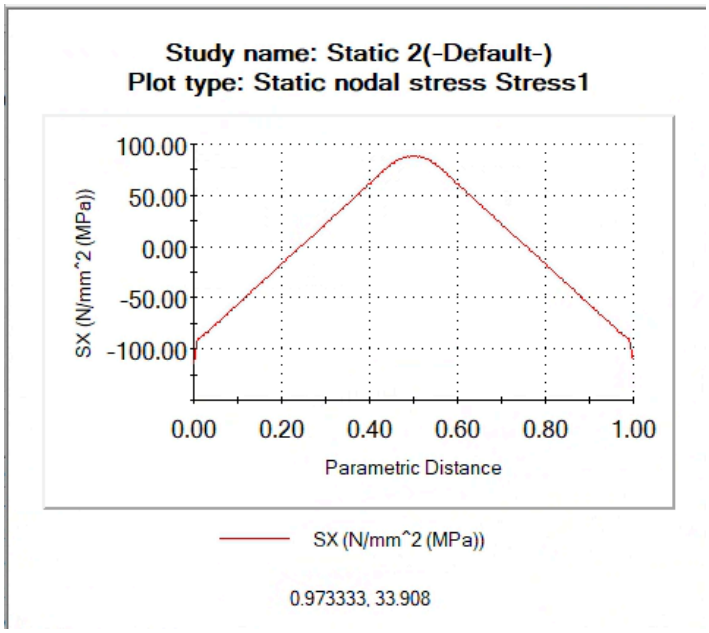
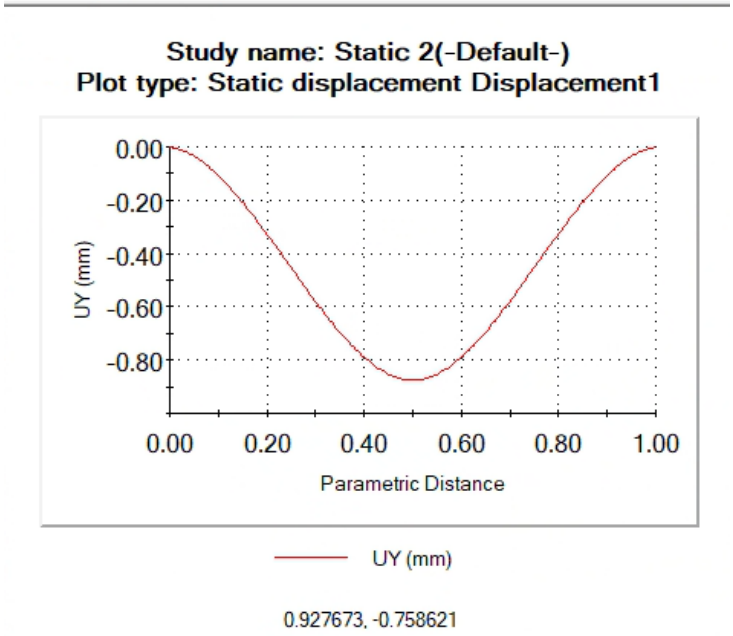
Raphael Lemus Solis

Deliverable 1



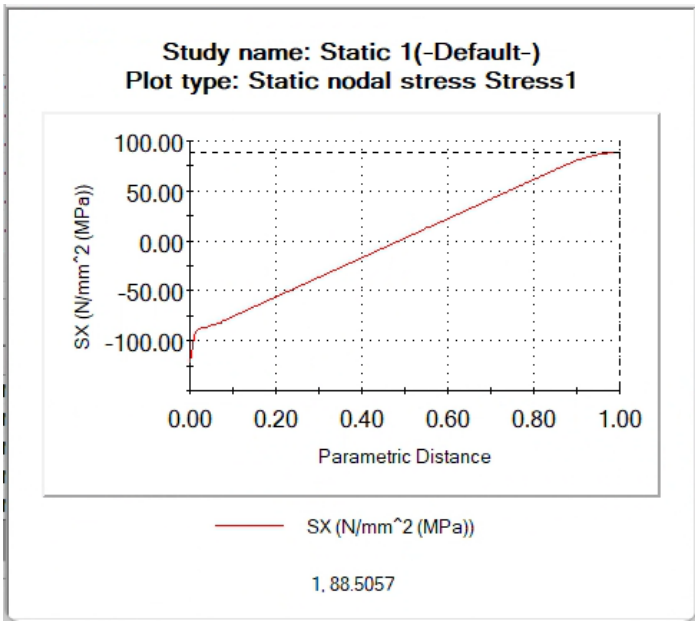
The peak tensile normal stress in the middle is 88.5 N/mm^2 (MPa), which is 88,500,000 N/m^2

Deliverable 2



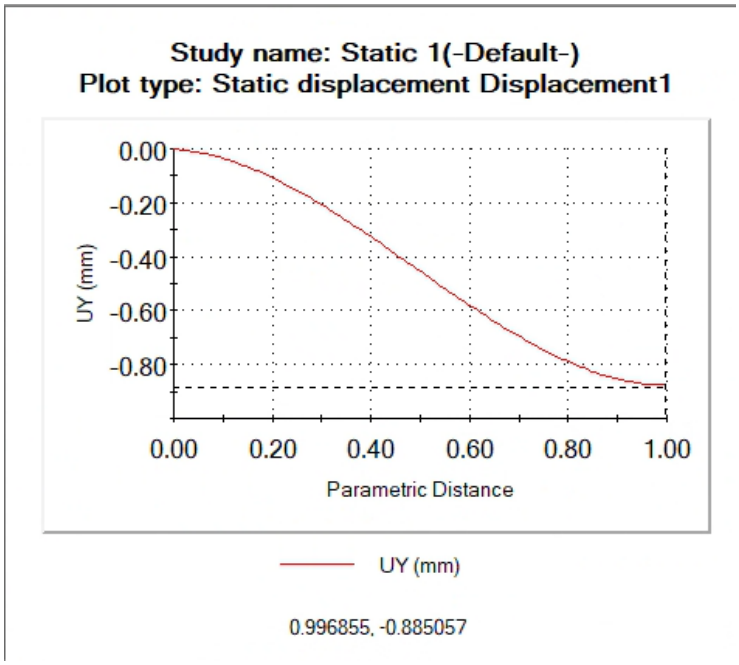
The maximum displacement along this line for the Y direction is 0.88 mm at the midpoint.

Deliverable 3



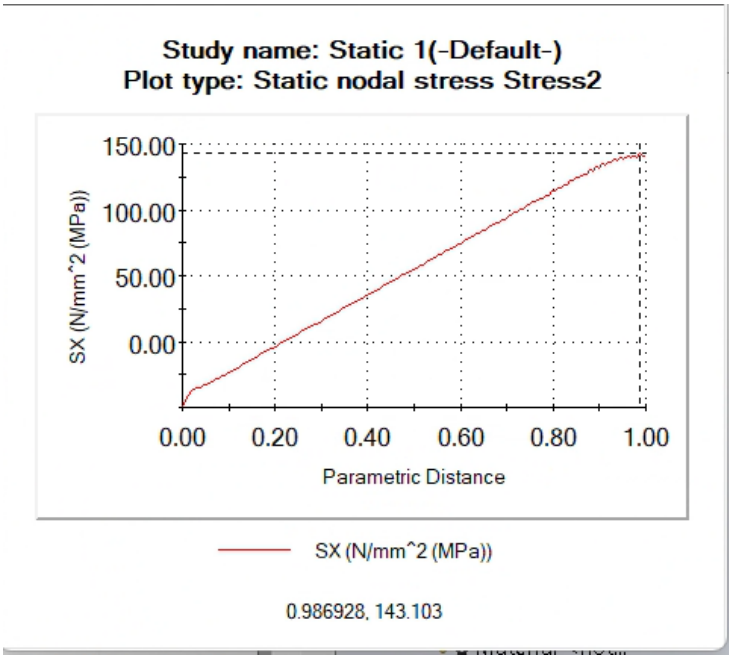
The peak tensile normal stress is 88.5 (N/mm^2)(MPa) or 88,500,000 N/m^2. Although it is a half-bar, the symmetry method will result in it being like the full pull-up bar. However, the one key difference is the parametric distance of the peak.

Deliverable 4



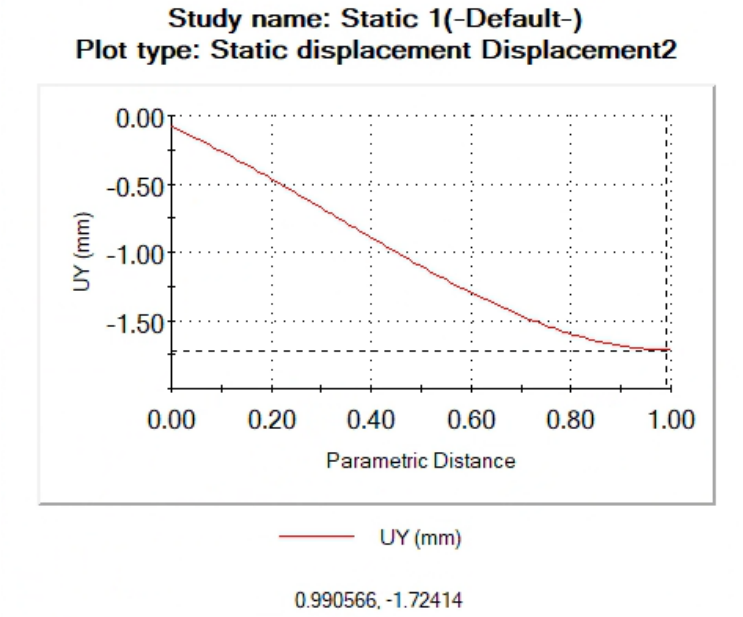
The Y displacement for the half-bar is close to the same as the Y displacement with the full bar. It had a displacement of about 0.885 mm. The key difference is the parametric distance where the displacement is located. Besides that, the symmetry method enables the half bar and full bar to be the same.

Deliverable 5



The peak tensile stress is halfway through the bar, and the value for the stress I got is 143.103 N/mm^2 (MPA) or 143,103,000 N/m^2. The tensile stress has increased by almost 2xs. However, I think this might be wrong because it should stay around the same/consistent. I believe this to be true because the full and half bars were the same. I might have done something different for the holes.

Deliverable 6



The peak displacement in the Y direction is halfway through the bar, and the value for the stress I got is 1.72 mm. The displacement that was achieved has increased by almost 2xs. However, I think this might be wrong because it should stay around the same/consistent. I believe this to be true because the full and half bars were the same. I might have done something different for the holes.